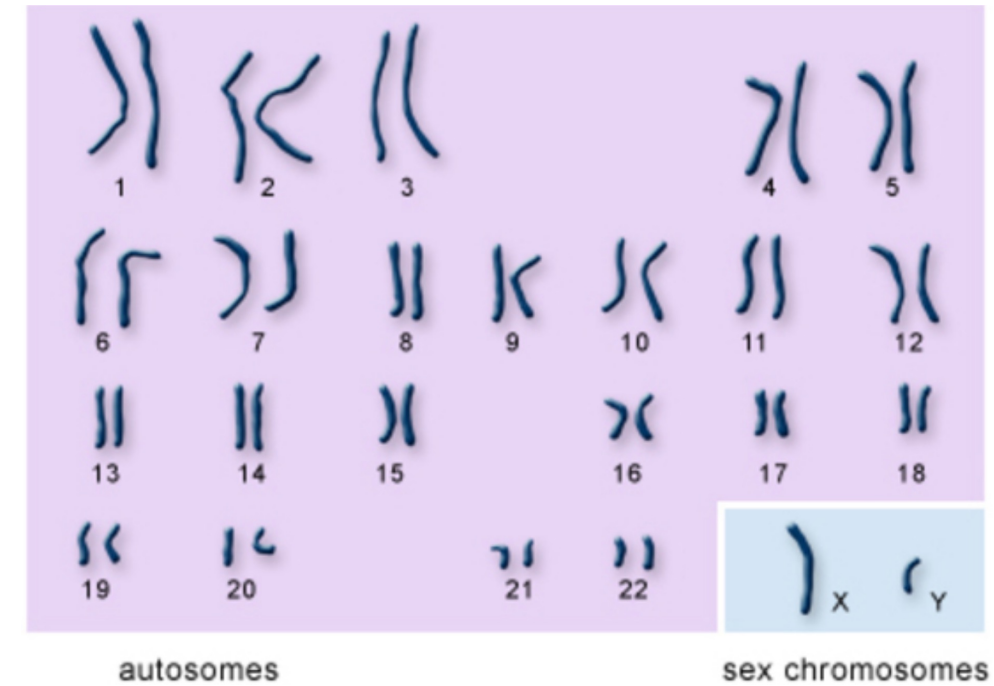
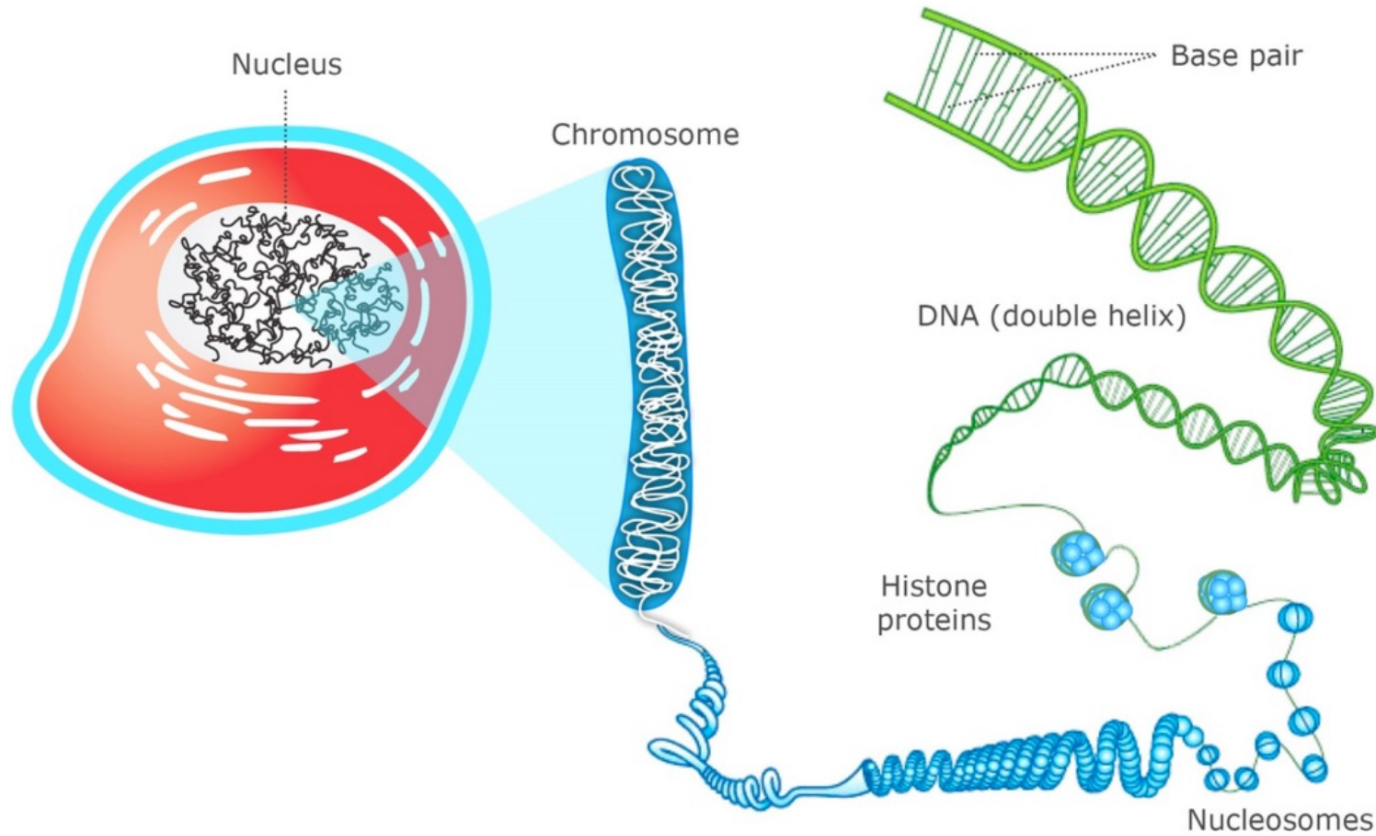


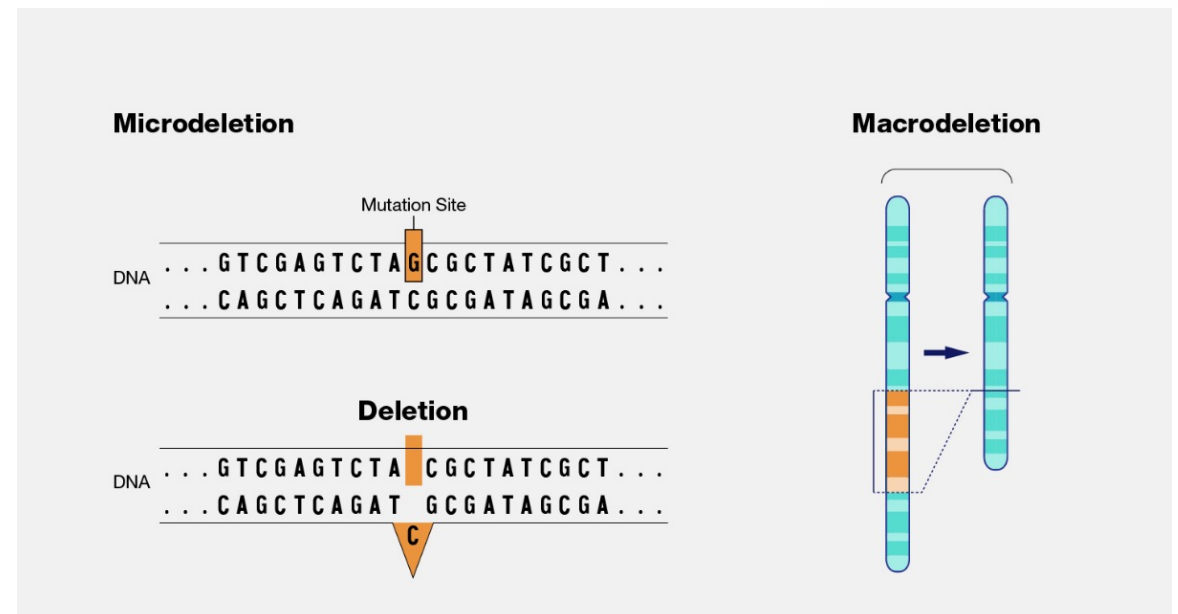
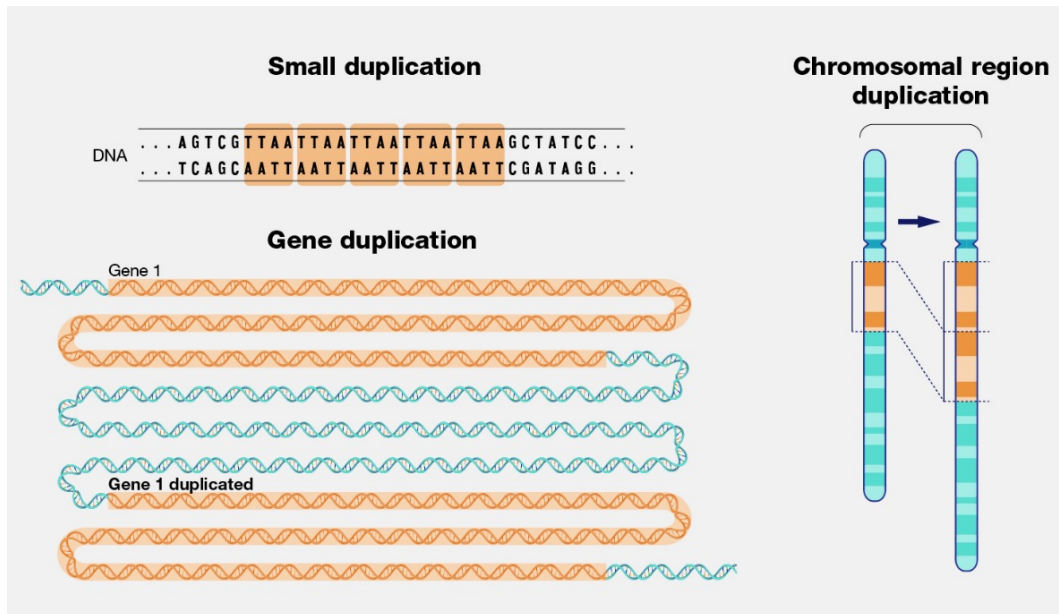
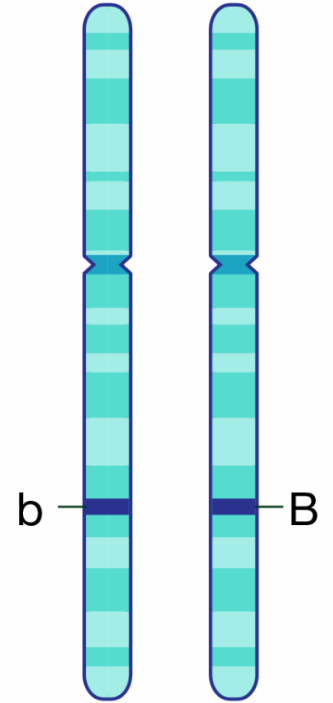
Detecting the cause of diseases by analyzing DNaseq data

Neuro-genomics class 12

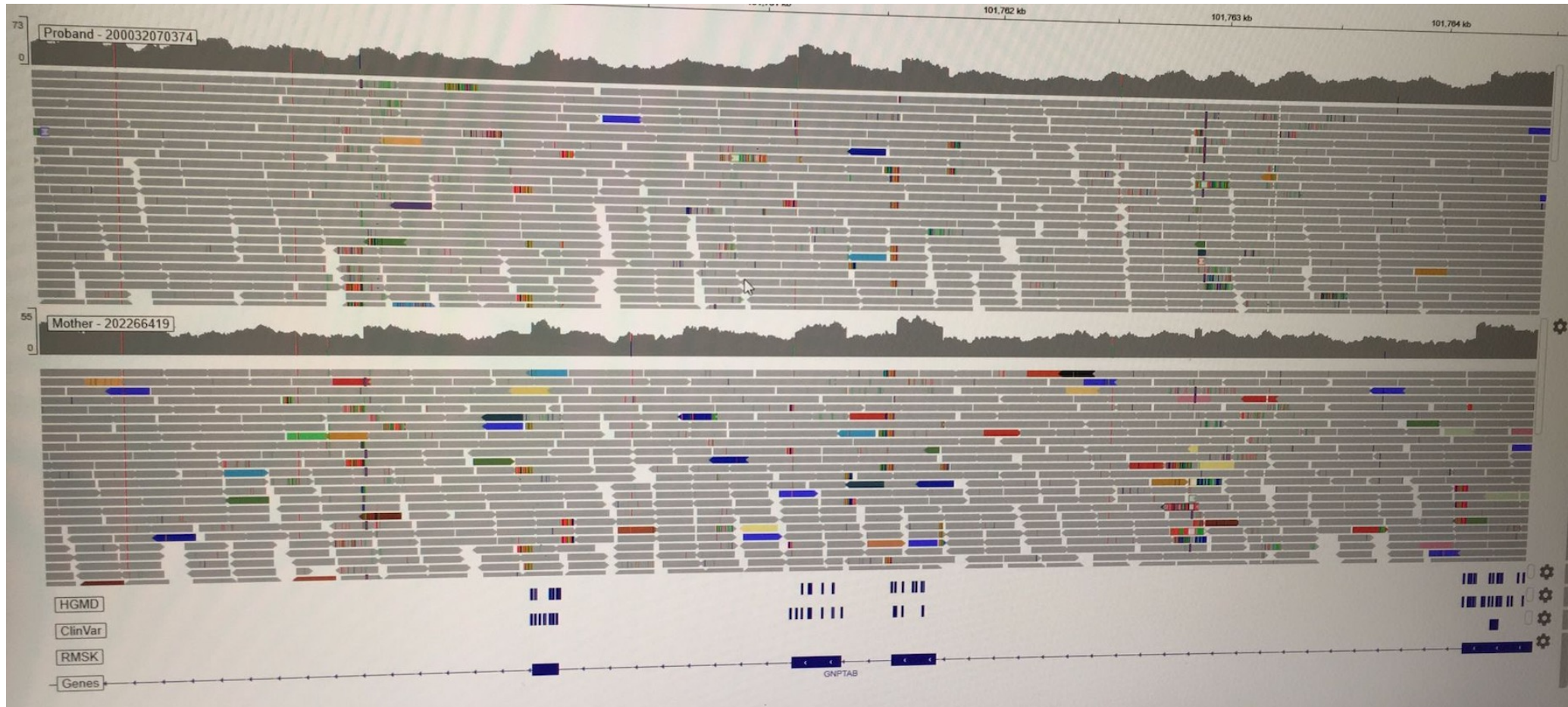
Reminder: human chromosomes



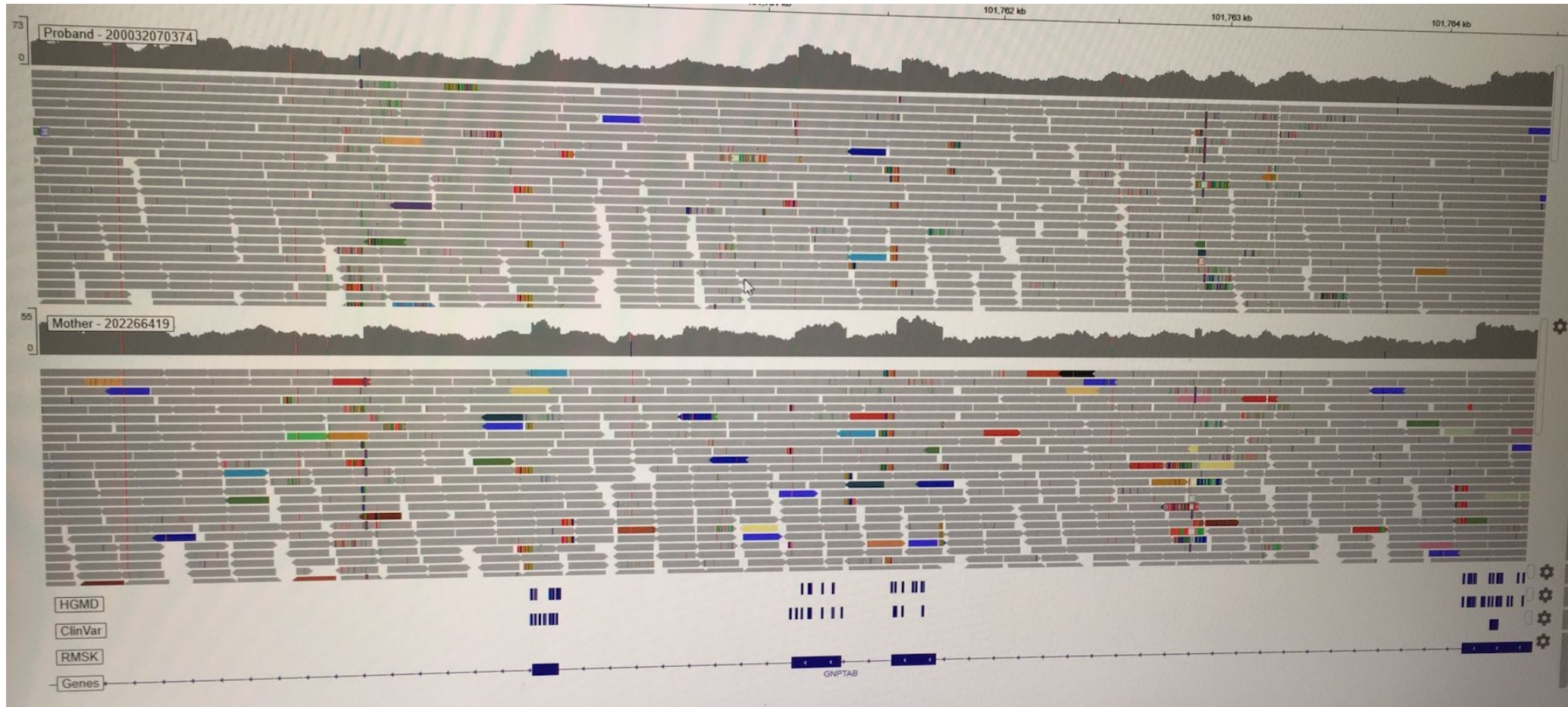
Reminders: Alleles, duplication, deletion



Warming up question: why in some cases DNaseq show more reads in exons compared to introns?

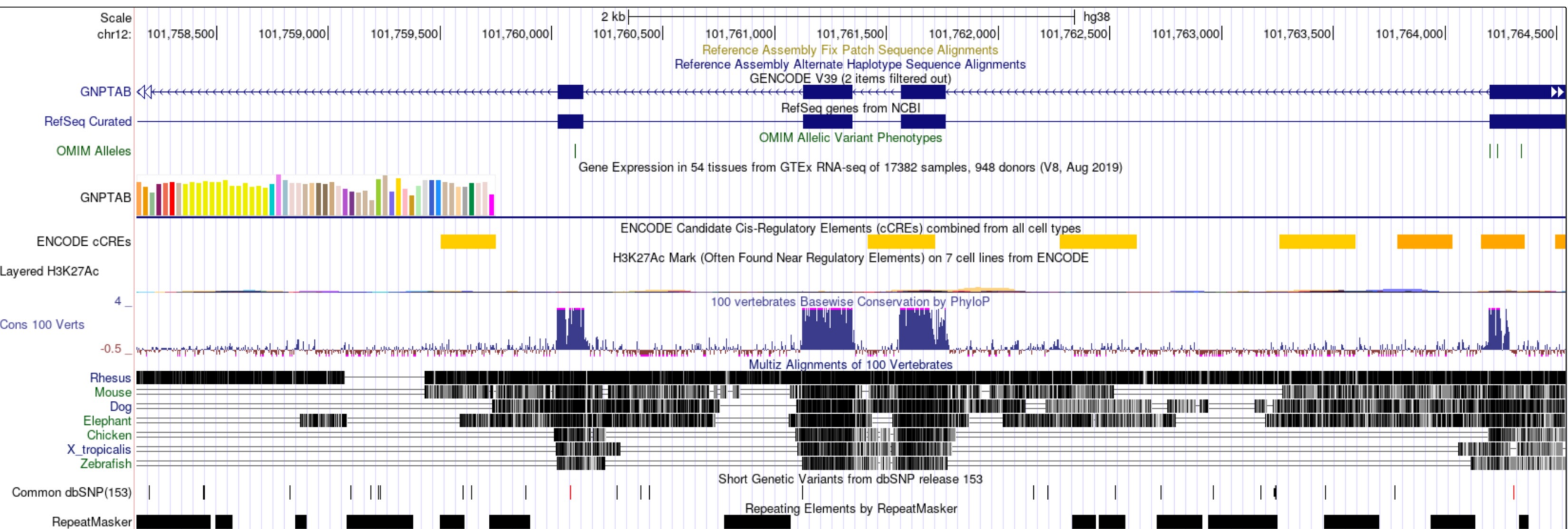


Warming up question: why in some cases DNaseq show more reads in exons compared to introns?

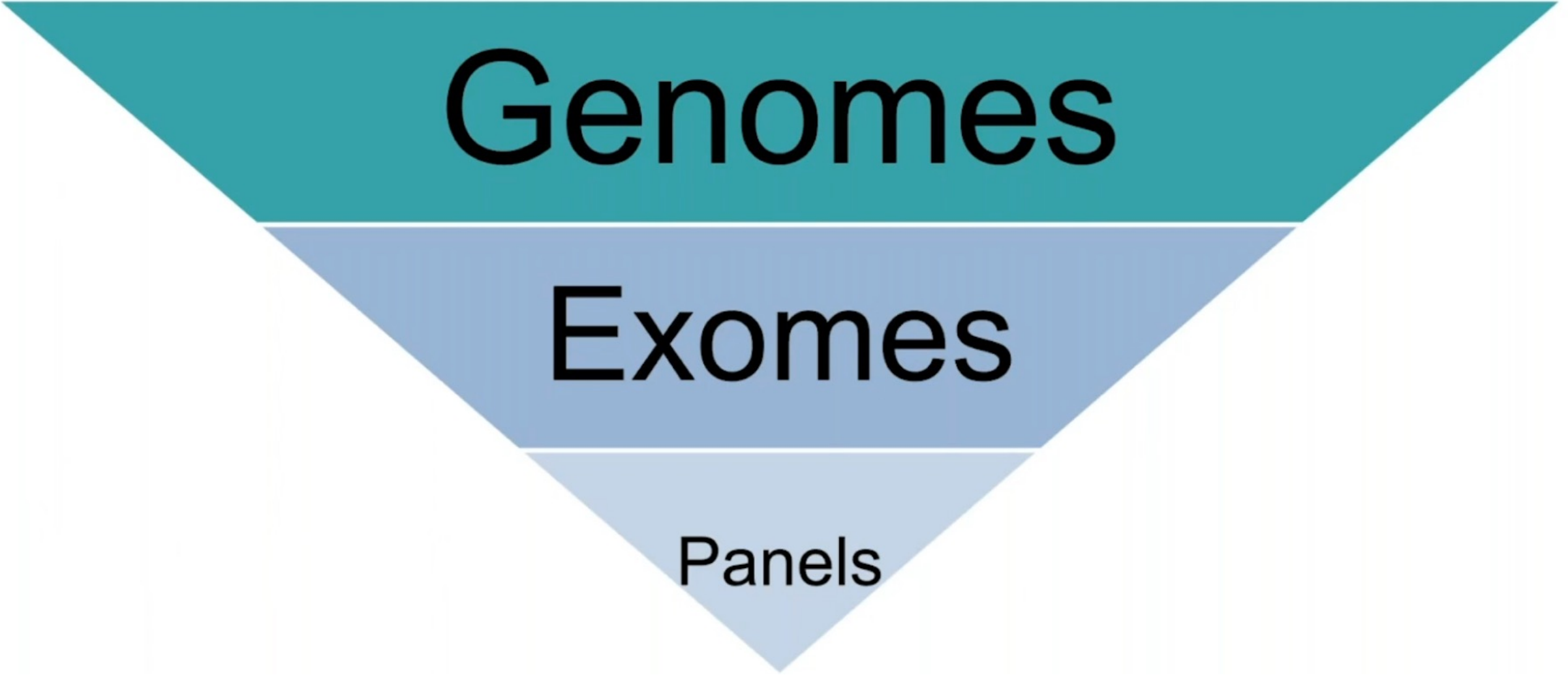


- Inserted processed transcripts via L1 retroelements

Warming up question: why in some cases DNase-seq show more reads in exons compared to introns?



- Mappability



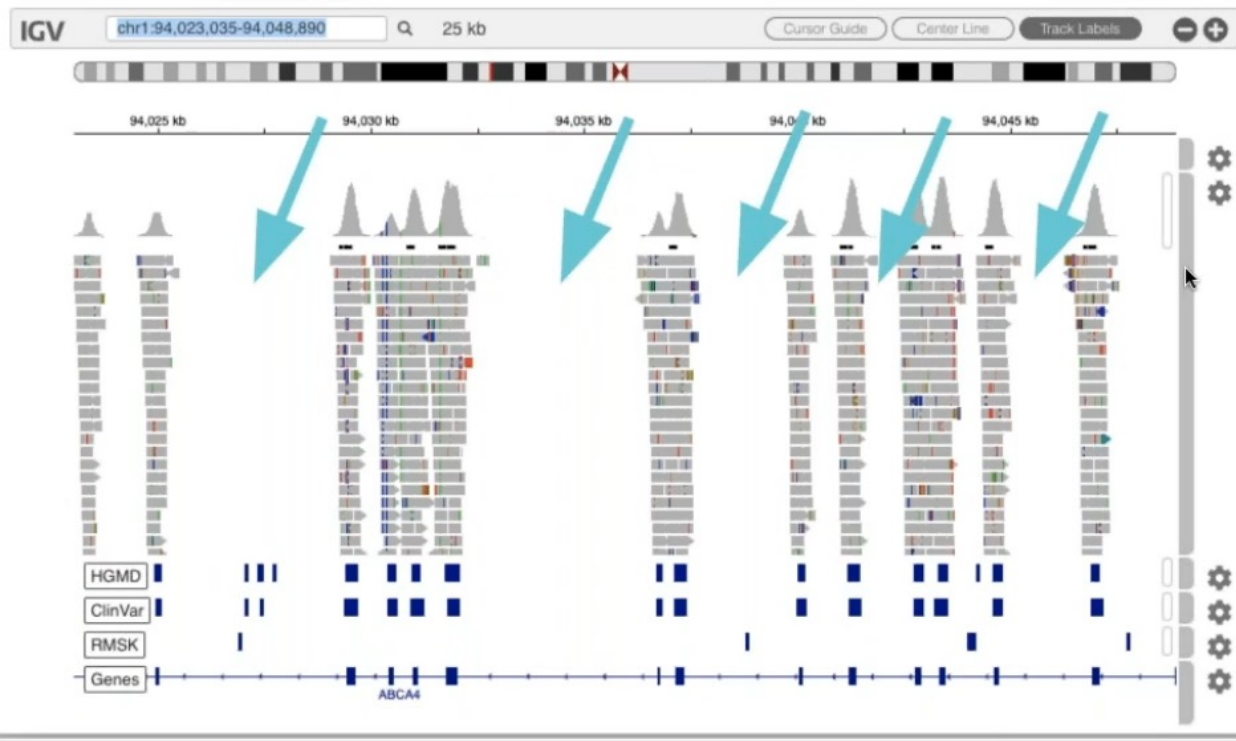
Genomes

Exomes

Panels

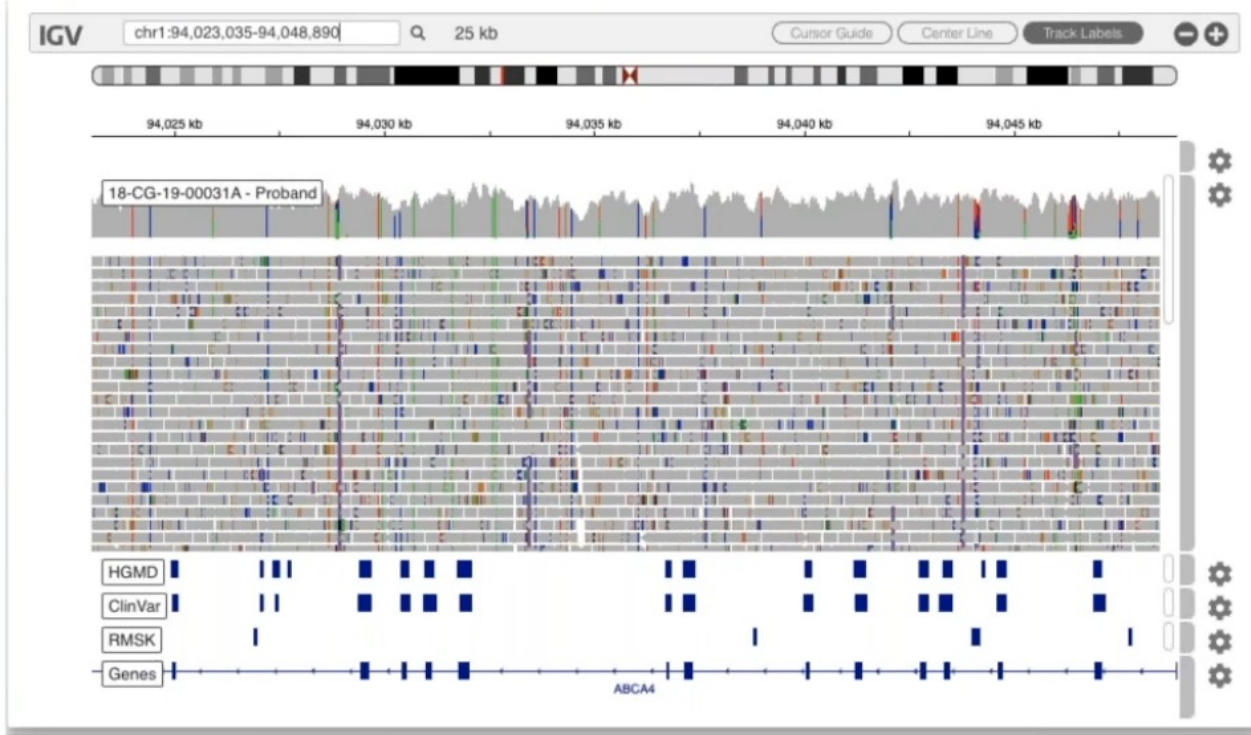
Exome/Panel

1-2% of all DNA, has data holes



Genome

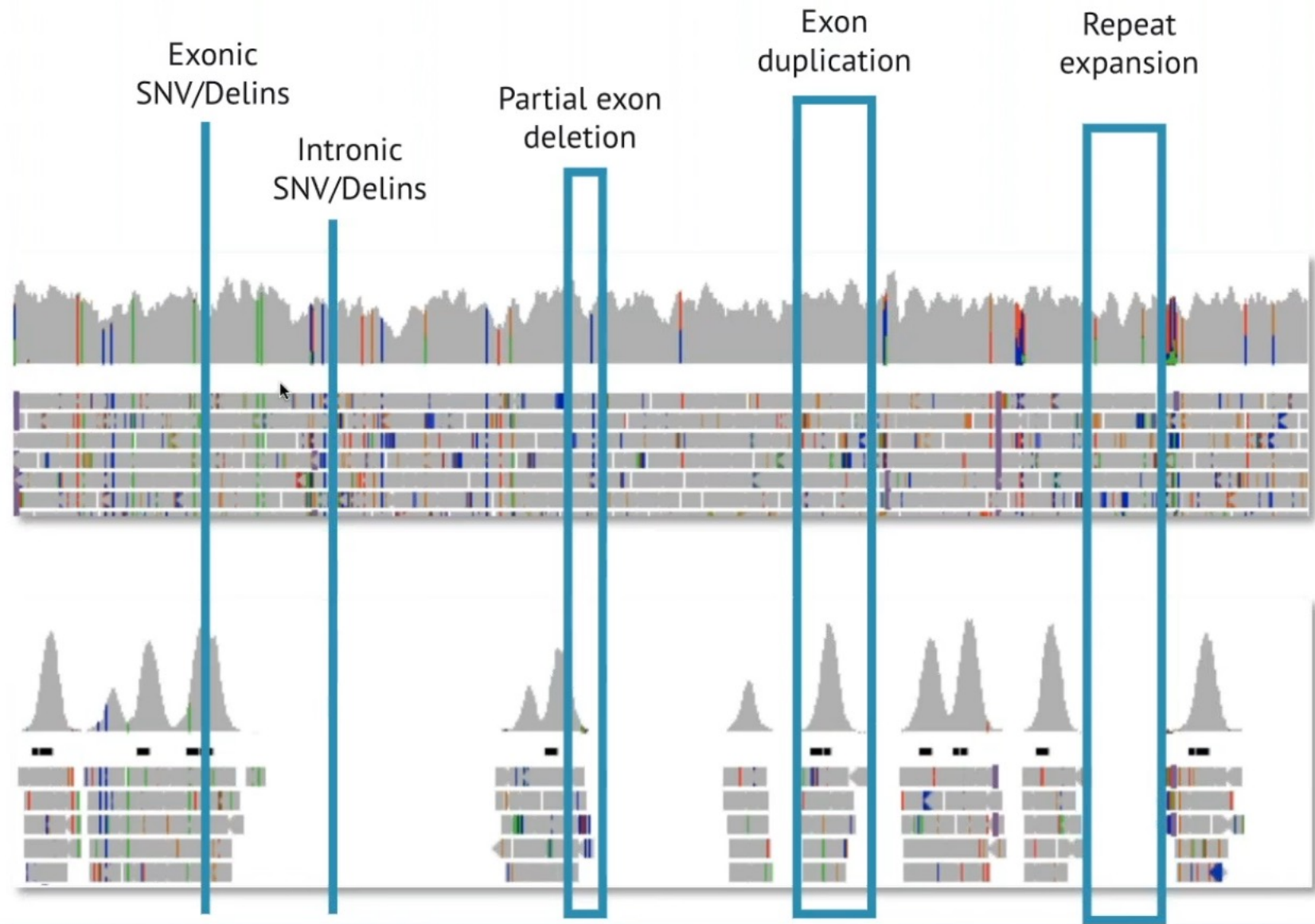
98% of all DNA, has no data holes



SAME SCALE

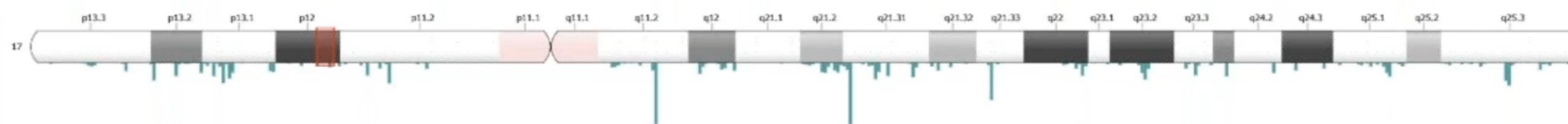
Genomes

Exomes



Multiple models uncover structural variants

Chromosome 17 Duplication



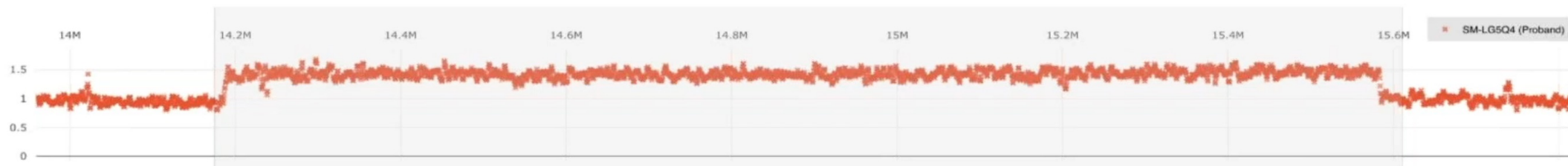
VAFs



Split/Discordant Reads



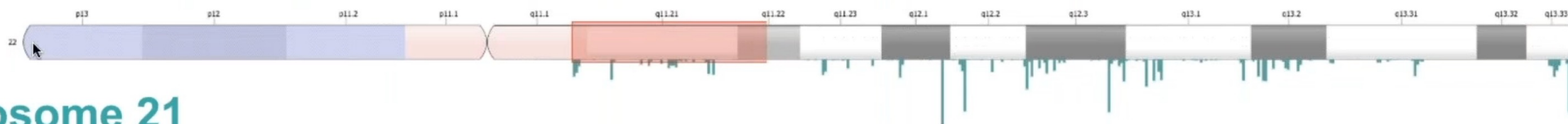
Copy Counts



- VAF = variant allele frequency. VAF is the percentage of sequence reads observed matching a specific DNA variant divided by the overall coverage at that locus.

Multiple models uncover structural variants

Chromosome 21 Deletion



VAFs



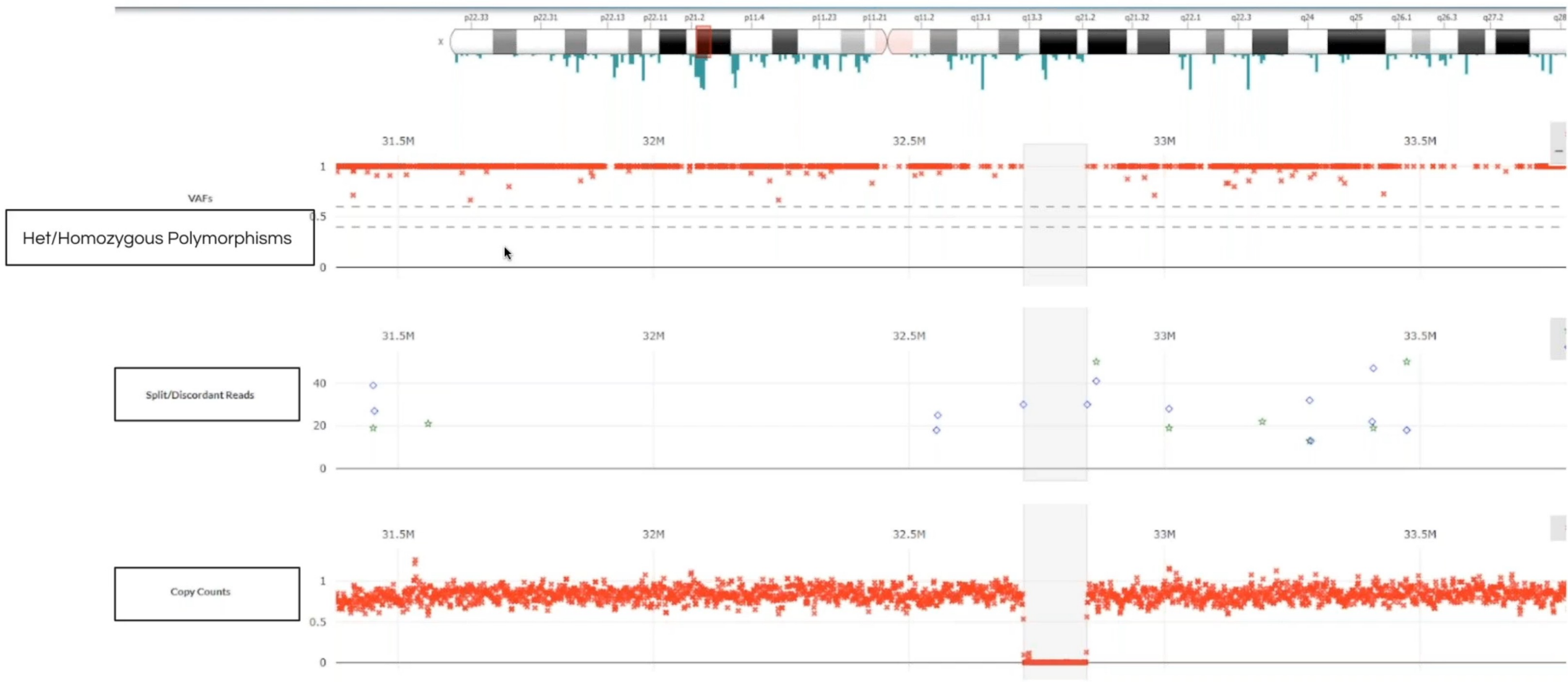
Split/Discordant Reads



Copy Counts



Multiple models uncover structural variants





A typical patient with rare disease

6+

Takes 6+ years
to be diagnosed

17



Visits 17 physicians



Spends an average
of \$20,000

Genome diagnostic rates

Specialty	Genome Detection Rate
General Genetics Clinic	68.3% (Stark et al, 2016)
Intellectual Disability	27% (Lindstrand et al 2019)
Congenital Anomalies/ Intellectual Disabilities	38% (Manickam et. al, 2021)
Neurodevelopmental	47% (Soden et al, 2018)
Neonatology	43% (Farnaes et al, 2018)
Congenital Heart Disease	31% (Alankarage et. al, 2019)
Epilepsy	63% (Palmer et. al, 2021)
Pediatric Mitochondrial Disorders	55% (Riley et al, 2020)

Case 1: phenotypes



39 year old
female

Juvenile Onset

Progressive gait and
balance difficulty

Paraplegia/paraparesis

Coordination difficulty

Dysarthria

Positive Babinski sign

Paresthesia of distal
lower extremities

chr9q21.11q21.11x1(69,071,296-69,074,068) [Pathogenic]

Primary ***

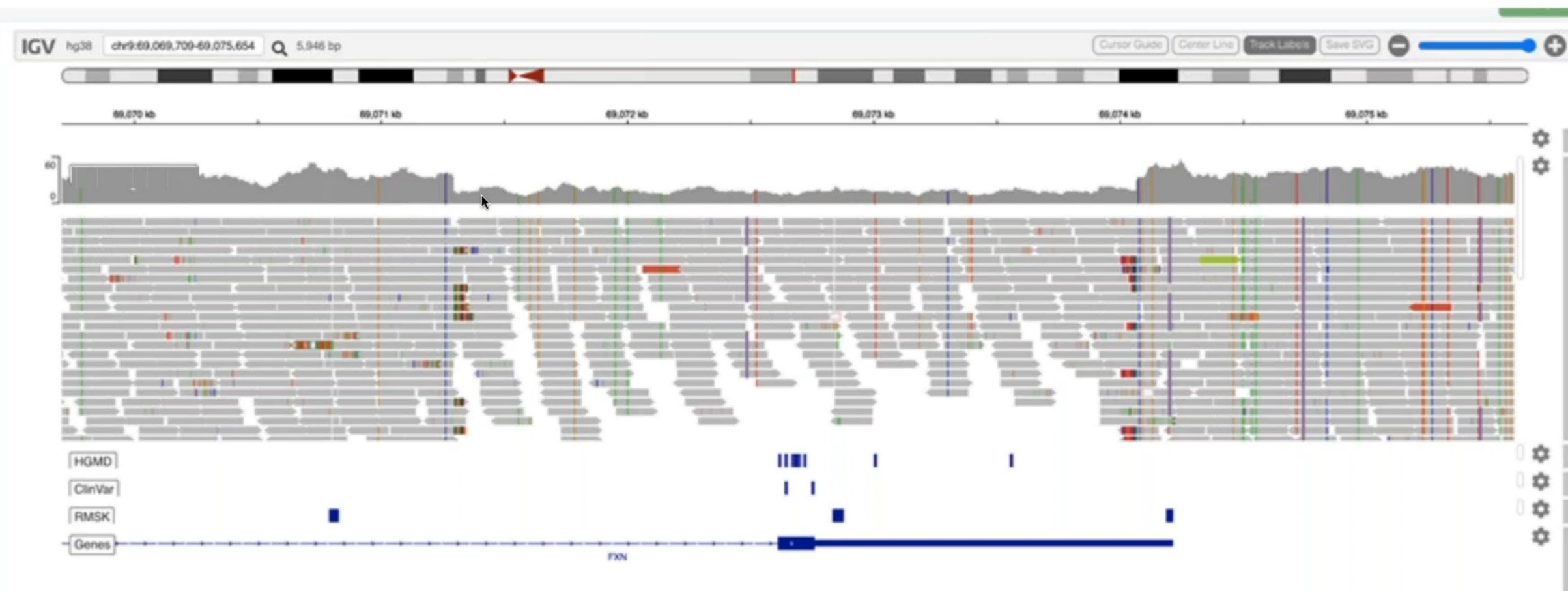
This deletion impacts the last exon of FXN. Deletions of this exon have been reported in the literature in individuals with Friedreich ataxia (PMID: 22409940).

2,772 bp

[HGMD]:Friedreich ataxia; Friedreich ataxia, early-onset



Deletion	Exonic	Heterozygous	7 44	0	1 (0%)	N/A	N/A	213,231	3	2
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FXN encodes a mitochondrial protein which is regulating mitochondrial iron transport and respiration. Diseases associated with FXN include Ataxia

Case 2: phenotypes



1 year old female

7 month diagnostic odyssey

FTT

Low muscle tone

Liver
transaminases
elevation

Gross motor delay

Frontal upsweep,
slightly wide-spaced
eyes, prominent
nose

Recurrent
Infections

Scheduled for a liver
transplant

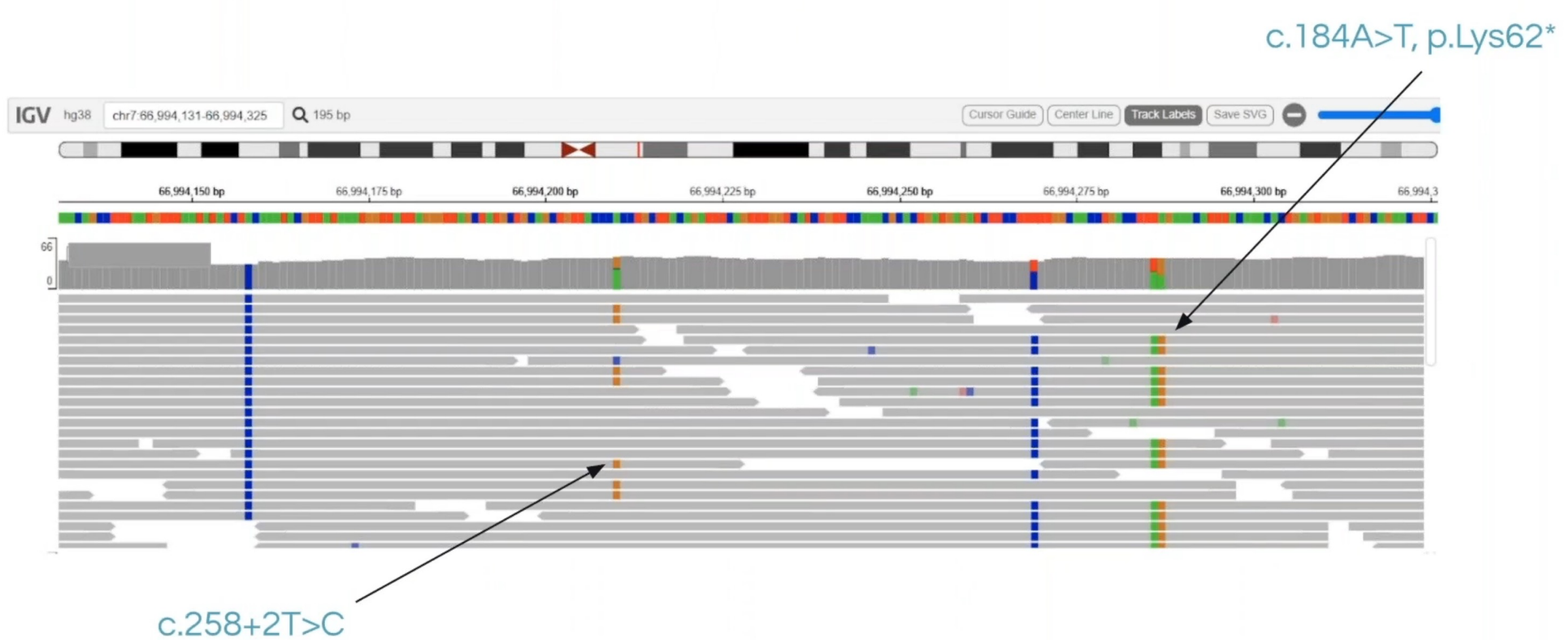
Normal Biochemical
Testing

Normal
Endocrinology

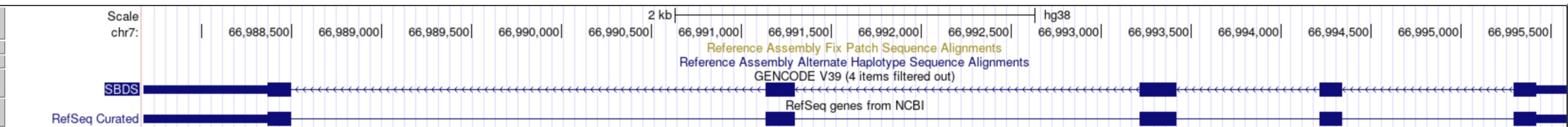
SBDS gene plays an essential role in ribosome biogenesis

Nonsense variant = a point mutation in a sequence of DNA that results in a premature stop codon

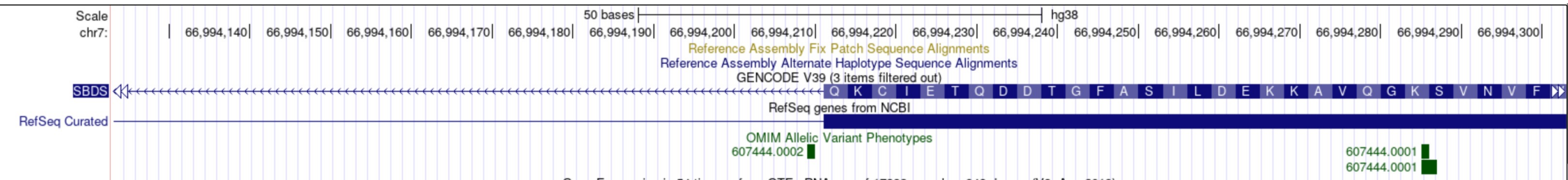
Splice site variant = a point mutation at the boundary of an exon and an intron which can disrupt RNA splicing



SBDS - overall



SBDS – exon #3

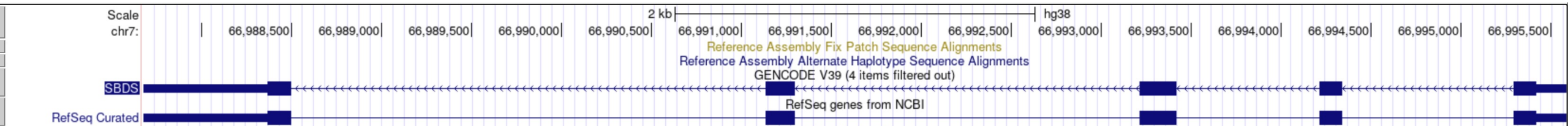


OMIM Allelic Variant: [607444.0001](#) SHWACHMAN-DIAMOND SYNDROME 1
OMIM: [607444](#): SBDS ribosome maturation factor
Amino Acid Replacement: LYS62TER

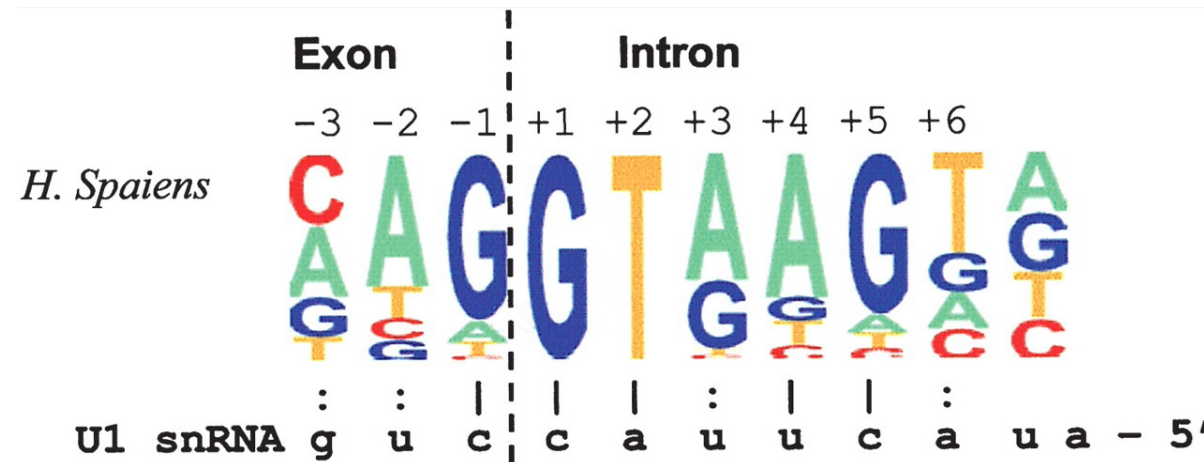
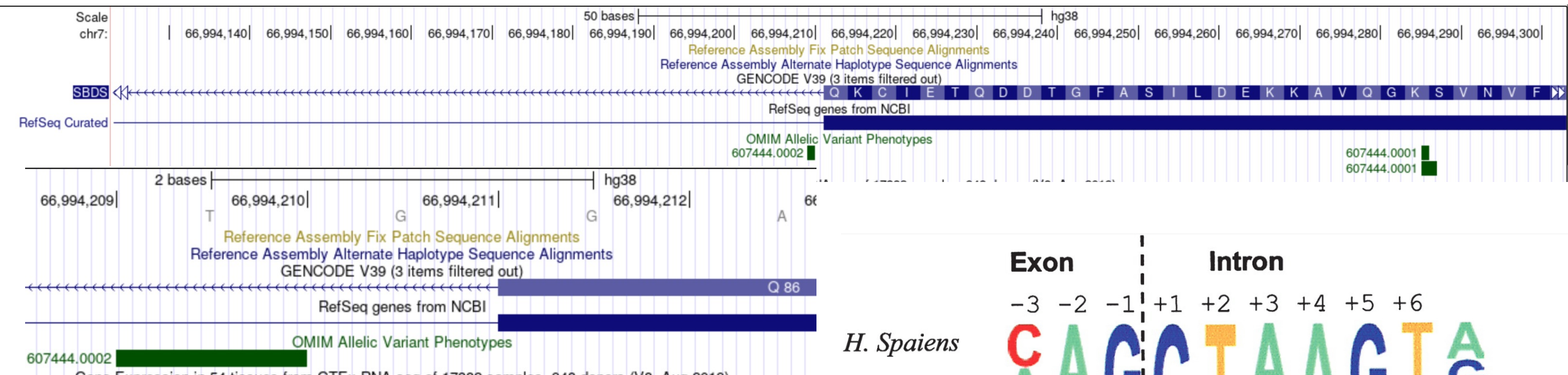
OMIM = a catalog of human genes and genetic phenotypes

Shwachman-Diamond syndrome is an inherited condition that affects many parts of the body, particularly the bone marrow, pancreas, and bones.

SBDS - overall



SBDS – exon #3



OMIM Allelic Variant: [607444.0002](#) SHWACHMAN-DIAMOND SYNDROME 1
OMIM: [607444](#): SBDS ribosome maturation factor
Amino Acid Replacement: IVS2DS, T-C, +2

These mutations were not detected using exome sequencing, due to mismapping



ORIGINAL ARTICLE

Shortfall of exome analysis for diagnosis of Shwachman-Diamond syndrome: Mismapping due to the pseudogene *SBDSP1*

Mamiko Yamada, Tomoko Uehara, Hisato Suzuki, Toshiki Takenouchi, Ayano Inui, Masako Ikemiyagi, Isamu Kamimaki, Kenjiro Kosaki ✉

First published: 15 May 2020 | <https://doi.org/10.1002/ajmg.a.61598>

Case 3: phenotypes



2 year old male

2 year diagnostic odyssey

Microcephaly

Heat Intolerance

Global Developmental Delay

Seizures
(onset 1 mo old)

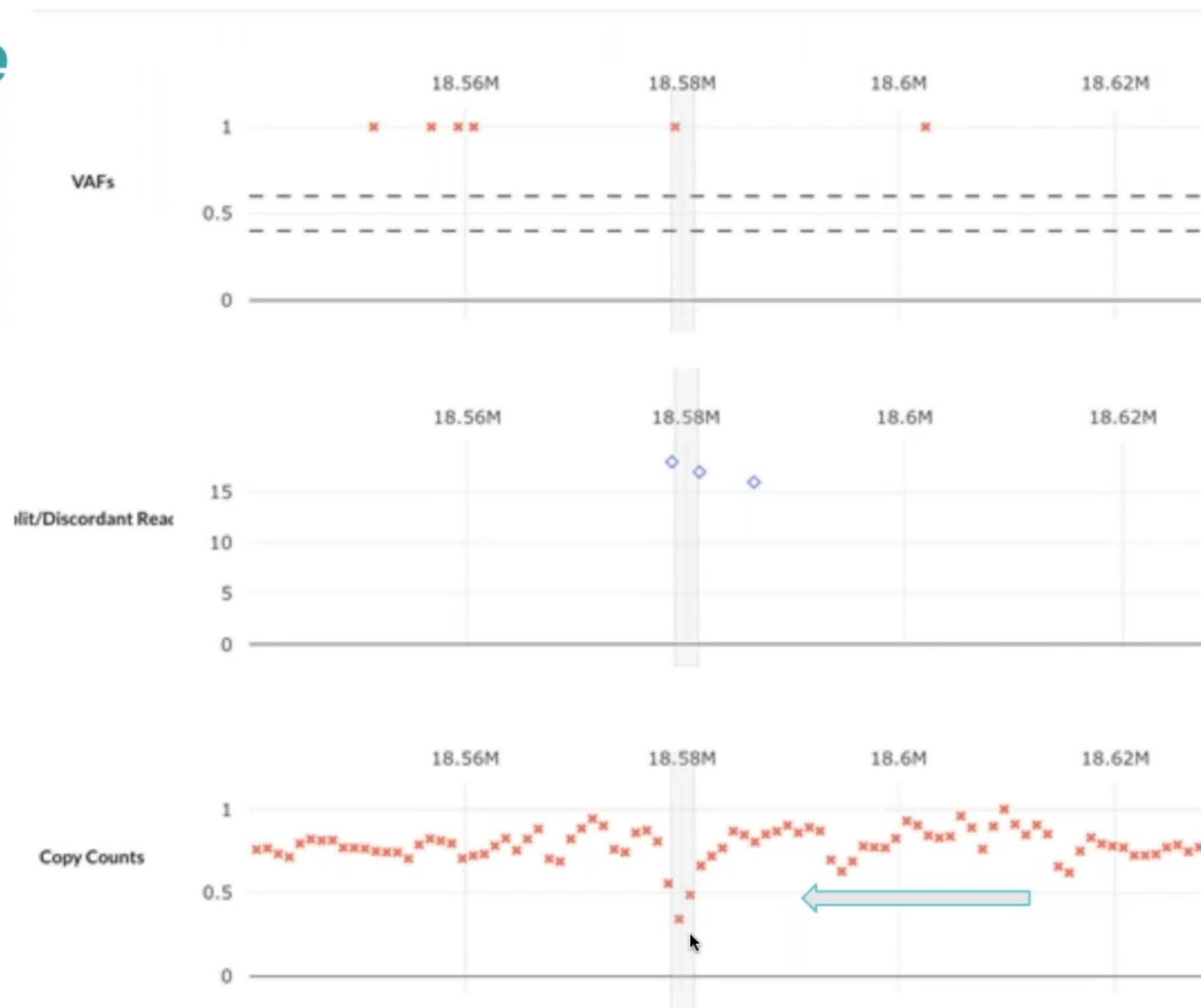
Vision Abnormality

Hypotonia

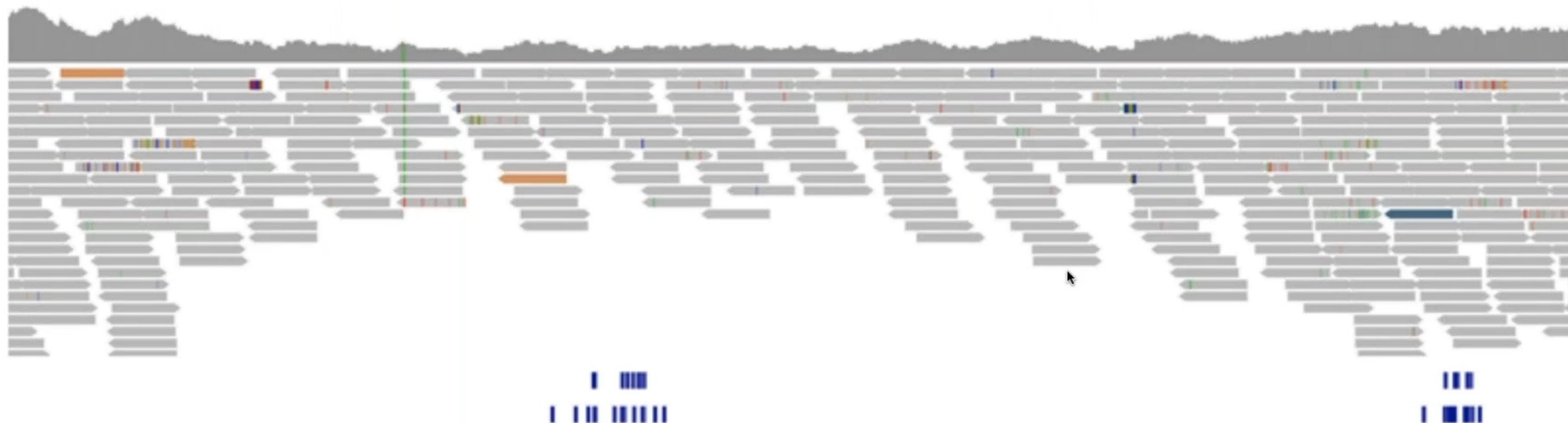
Dysphagia
(tube feeding)

Neurodevelopmental Delay

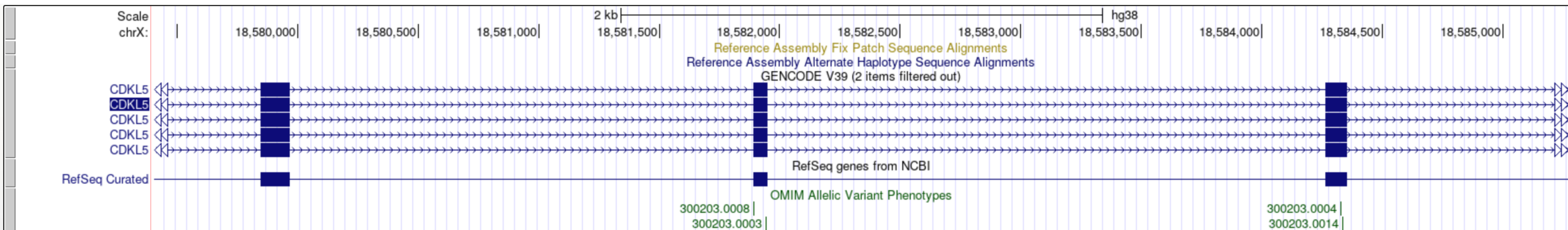
CDKL5 exon 7 *de novo* mosaic pathogenic deletion



CDKL5 single exon deletion - Mosaic - X-linked



Surrounding sequence is intronic; not captured by exomes



OMIM Allelic Variant: [300203.0008](#) DEVELOPMENTAL AND EPILEPTIC ENCEPHALOPATHY 2
OMIM: [300203](#): Cyclin-dependent kinase-like 5 (serine/threonine protein kinase 9)

Developmental and Epileptic Encephalopathy (DEE) refers to a group of severe epilepsies that are characterized both by seizures, which are often drug-resistant, as well as encephalopathy, which is a term used to describe significant developmental delay or even loss of developmental skills.

Case 4: phenotypes



14 year old
male

11 year
diagnostic
odyssey

Muscle weakness

Ataxia

Tremor

Paresthesia

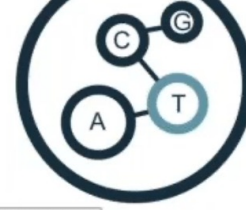
Mild waddling gait

Intermittent head
titubation

Normal CK

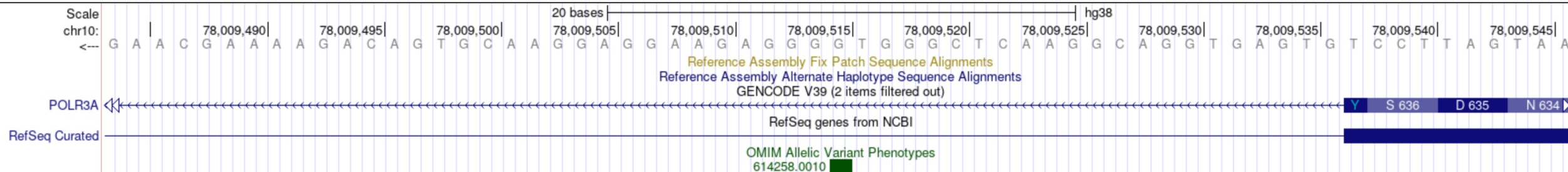
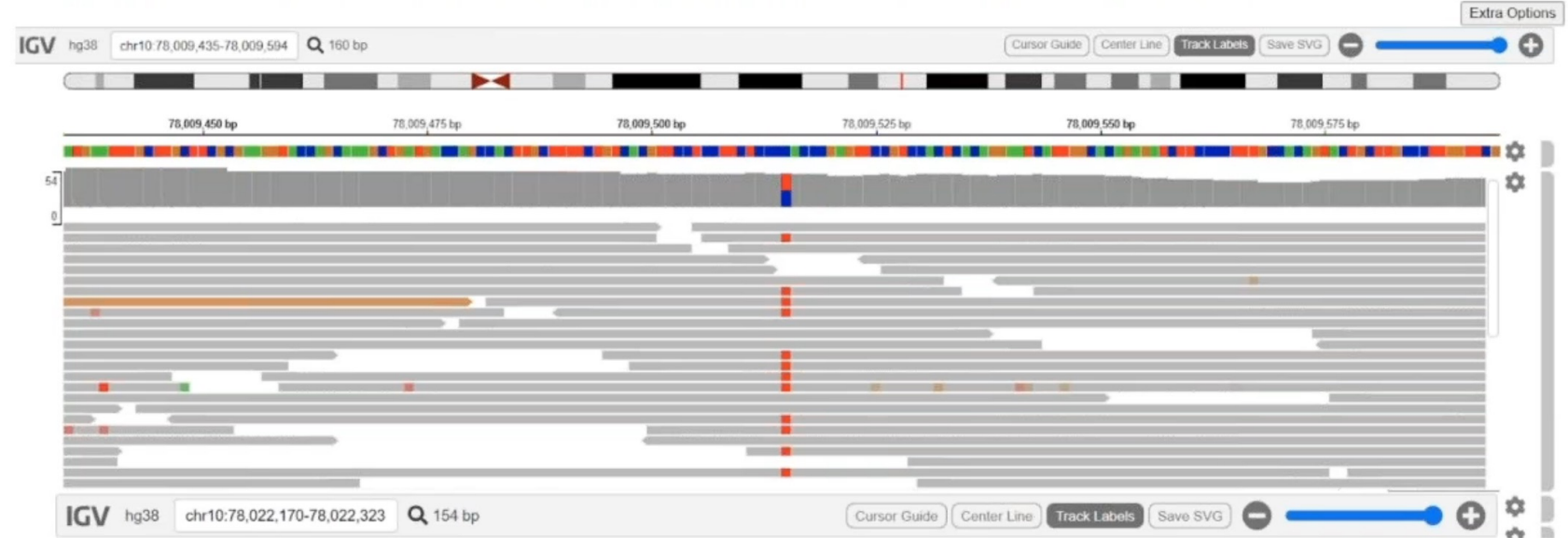
Normal Brain MRI

Normal EKG



POLR3A gene - compound heterozygous variants

c.1909+22G>A



- Mutations in intron regions can have an effect, even if they are not in splice site
- Much more DNaseq data is needed to detect these important regions inside introns



***POLR3A*-related spastic ataxia: new mutations and a look into the phenotype**

Jon Infante¹ · Karla M. Serrano-Cárdenas¹ · Marc Corral-Juan² · Xavier Farré² · Ivelisse Sánchez² · Enrique M. de Lucas³ · Antonio García⁴ · José Luis Martín-Gurpegui¹ · José Berciano¹ · Antoni Matilla-Dueñas²

Received: 1 July 2019 / Revised: 28 August 2019 / Accepted: 3 October 2019
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Abstract

Adolescent-onset spastic ataxia is a proposed novel phenotype in compound heterozygous carriers of an intronic mutation (c.1909 + 22G > A) in the *POLR3A* gene. Here, we present ten new cases of *POLR3A*-related spastic ataxia and discuss the genetic, clinical and imaging findings. Patients belonged to six pedigrees with hereditary spastic paraplegia or cerebellar ataxia of unknown origin. All affected subjects presented with compound heterozygous variants, comprising c.1909 + 22G > A in combination in each pedigree with one of the following novel mutations (Thr596Met, Tyr665LeufsTer11, Glu198Ter, c.646-687_1185 + 844del). The new mutations segregated with the phenotype in all families. The phenotype combined variable cerebellar ataxia, gait and lower limb spasticity, involvement of central sensory tracts and in some cases also intention tremor. The reportedly characteristic hyperintensity along the superior cerebellar peduncle on MRI was observed in ~80% of the cases. Our study extends the clinical and molecular phenotype further supporting the pathogenic role of the c.1909 + 22G4A intronic mutation and identifying four novel causative mutations in *POLR3A*-related spastic ataxia. Certain characteristic MRI features may be useful to guide genetic diagnosis.

Keywords Hereditary spastic paraplegia · Hereditary ataxia · Spastic ataxia · *POLR3A*

Spastic ataxia = a complex neurodegenerative disorder usually characterized by early childhood onset of cerebellar ataxia, pyramidal tract signs, and peripheral neuropathy. Most patients become wheelchair-bound; cognitive function is usually not affected